Data availability

The datasets generated and/or analysed during the current study are available in the Data source: https:// www. kaggle. com/ datas ets/ sanjo li02/ lung- cancer.
Received: 2 February 2024; Accepted: 27 March 2024

من هنا تحصل على الداتاسيت

The original paper that published dataset is:

Hong, Z.Q. and Yang, J.Y. "Optimal Discriminant Plane for a Small Number of Samples and Design Method of Classifier on the Plane", Pattern Recognition, Vol. 24, No. 4, pp. 317-324, 1991.

يجب قراءة الداتاسيت قبل اي خطوة

df=pd.read_csv('/survey lung cancer.csv')

Pre-processing

1) Data cleaning by:
   - removing duplicates ازالة الحالات المتشابهة

     #Checking for Duplicates

     df.duplicated().sum()

     #Removing Duplicates

     df=df.drop_duplicates()
   - converting object data type to numerical values (found GENDER and LUNG_CANCER attributes are object data type)

     هنا تحويل القيم للميزات الى شكل رقمي وهما العمودان المحددان بين الاقواس عند كتابة الايعاز ستلاحظ انواع البيانات

     df.info()

```
<class 'pandas.core.frame.DataFrame'>
Index: 276 entries, 0 to 283
Data columns (total 16 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   GENDER                 276 non-null    object
 1   AGE                    276 non-null    int64
 2   SMOKING                276 non-null    int64
 3   YELLOW_FINGERS         276 non-null    int64
 4   ANXIETY                276 non-null    int64
 5   PEER_PRESSURE          276 non-null    int64
 6   CHRONIC DISEASE        276 non-null    int64
 7   FATIGUE                276 non-null    int64
 8   ALLERGY                276 non-null    int64
 9   WHEEZING               276 non-null    int64
 10  ALCOHOL CONSUMING      276 non-null    int64
 11  COUGHING               276 non-null    int64
```

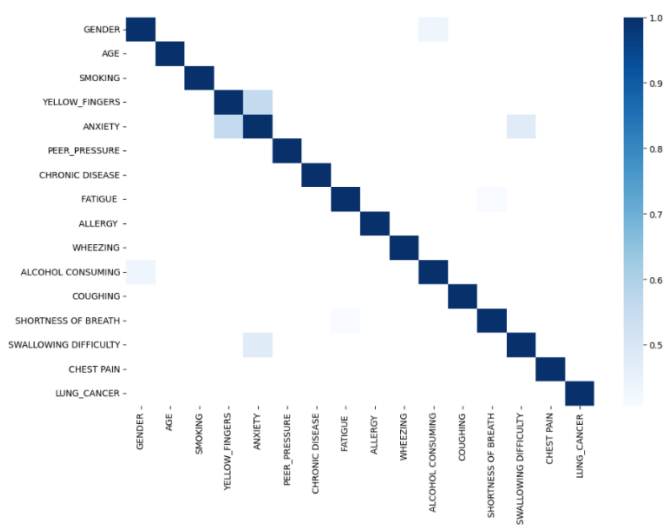
```
 12  SHORTNESS OF BREATH     276 non-null    int64
 13  SWALLOWING DIFFICULTY   276 non-null    int64
 14  CHEST PAIN              276 non-null    int64
 15  LUNG_CANCER             276 non-null    object
dtypes: int64(14), object(2)
```

```
from sklearn import preprocessing
le=preprocessing.LabelEncoder()
df['GENDER']=le.fit_transform(df['GENDER'])
df['LUNG_CANCER']=le.fit_transform(df['LUNG_CANCER'])
df['SMOKING']=le.fit_transform(df['SMOKING'])
df['YELLOW_FINGERS']=le.fit_transform(df['YELLOW_FINGERS'])
df['ANXIETY']=le.fit_transform(df['ANXIETY'])
df['PEER_PRESSURE']=le.fit_transform(df['PEER_PRESSURE'])
df['CHRONIC DISEASE']=le.fit_transform(df['CHRONIC DISEASE'])
df['FATIGUE ']=le.fit_transform(df['FATIGUE '])
df['ALLERGY ']=le.fit_transform(df['ALLERGY '])
df['WHEEZING']=le.fit_transform(df['WHEEZING'])
df['ALCOHOL CONSUMING']=le.fit_transform(df['ALCOHOL CONSUMING'])
df['COUGHING']=le.fit_transform(df['COUGHING'])
df['SHORTNESS OF BREATH']=le.fit_transform(df['SHORTNESS OF BREATH'])
df['SWALLOWING DIFFICULTY']=le.fit_transform(df['SWALLOWING DIFFICULTY'])
df['CHEST PAIN']=le.fit_transform(df['CHEST PAIN'])
df['LUNG_CANCER']=le.fit_transform(df['LUNG_CANCER'])
```

2) Feature engineering by:

- using **Pearson's Correlation matrix** (as a heat map) : (you will find that ANXIETY and YELLOW_FINGERS are high correlated, so that we create a new feature(attribute) combining them and added it to the dataset.

في الاسفل البيرسون ماتركس المفروض هكذا تنتج لك , لاحظ الاعمدة المتشابهة ملونة بالازرق الفاتح , سنقوم باستخدامها لانتج فيجر او ميزة جديدة نسميها باسم جديد

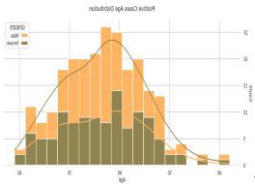


*df['ANX_YEL_FIN']=df['ANXIETY']*df['YELLOW_FINGERS']*

3) Data analysis based on positive and negative cases among males and females over the age distribution:

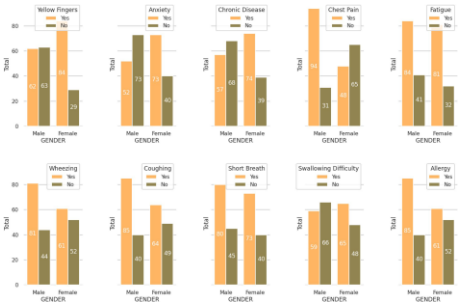
من خلال البيرسون ماتركس نلاحظ ان الجنس يشكل العامل الاكثر تاثيرا يليه لذلك نقوم بتحليل البيانات بناءً على الحالات الإيجابية والسلبية بين الذكور والإناث حسب الفئة العمرية و يشمل:

1.Positive case distribution age-wise over gender.



المفروض هكذا تحصل عليه

2.Distribution of positive and negative cases gender-wise over patient's symptoms

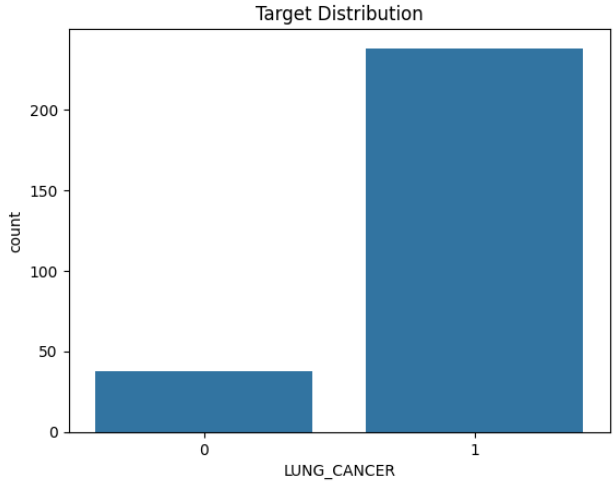


المفروض هكذا تحصل عليه

4) Check the distribution of Target variable.

- نلاحظ هنا ان الداتاسيت غير متوازنة لذلك يجب نستخدم تقنية لموازنة الفئات

```
sns.countplot(x='LUNG_CANCER', data=df,)
plt.title('Target Distribution');
```



5) Splitting independent and dependent variables

هذا يتم باعتبار العمود الاخير هو الجزء الذي يجب ان نتنبأ به

```
X = df.drop('LUNG_CANCER', axis = 1)
y = df['LUNG_CANCER']
```

6) The Data Split: by setting 80% or 70 % for training and 20% or 30% for testing.

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test= train_test_split(X, y, test_size= 0.20 or 0.30)
```

7) Imbalanced handling: ADASYN-Adaptive Synthetic Sampling was used to balance the dataset.

هنا نستخدم خوارزمية ادايسون لموازنة الفئات

```
from imblearn.over_sampling import ADASYN
adasyn = ADASYN(random_state=88)
X_train, y_train = adasyn.fit_resample(X_train, y_train )
```

8) Standardization: calculates the mean and standard deviation for each feature (attribute), and then transforms independent X by applying the standardization formula: $Z=(x-\mu)/\sigma$.

هنا نعمل ستاندرايزيشن للبيانات باستخدام طريقة z

```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X)
```

9) Dimensionality Reduction by: PCA Principal Component Analysis to reduce no of features and preserve the most important information only, that represents **75%** of the total variance in the input data.

هذه التقنية تستخدم لازالة الميزات غير الضرورية التي لا تسهم في صنع القرار

```
pca = PCA(n_components=0.75) #(retain 75% of variance)
X_pca = pca.fit_transform(X_scaled)
```

10) Then train models and assess the model

K-Nearest Neighbors (KNN), Random Forest (RF), XGBoost (XGB), and Gradient Boosting (GB).

Employ a stacking ensemble to leverage the complementary strengths of multiple classifiers. Stacking combines [K-Nearest Neighbors (KNN), Random Forest (RF), XGBoost (XGB), and Gradient Boosting (GB)] ; then compare results of stacking with other models.